(ô), grave **(ò)** and double grave **(ȍ)** accent indicates different types of tone accent. The letters **(ь)** and **(ъ)** stand for very short [i] and [u] sounds, **l'** denotes the palatalized version of l, and **ž** denotes a sound that was probably something like [ʒ]. Along with the approximate meanings of the words in this problem, case (NOM/GEN/LOC stands for nominative/genitive/locative) and number (SG/PL stands for singular/plural) for each word form.

Problem 11. Tocharian (2024, MSK)

Some verb forms of two closely related languages, Tocharian A and Tocharian B, and their English translations are given:

Tocharian A	**English**	**Tocharian B**	**English**
! pälskāt	he thought	! pälkānt	they saw
musnāl	he must climb	kotnatsi	to split
! skenāl	he must strive	paikāte	he wrote
kropnal	he must gather	käryātai	you (sg.) bought
! wenaträ	he drops	kraupātai	you (sg.) gathered
rināträ	he desires	kärsnāträ	he knows
kälpāte	you (sg.) received	prutknāträ	he becomes full
wärpnāmträ	we feel	mrausknāträ	he feels disgust for the world

(a) Translate the following verb forms into English, indicating which language each belongs to. If multiple variants are possible, give them all: **kälpnāl, mrauskāte, wärpāte, pekat, kropat, musāt, käryāmte**

(b) There are small differences between the different Tocharian A manuscripts. For example, in one of the manuscripts the dialectal form **kropnasi** was found. Indicate how this form should look in "standard" Tocharian A, and translate it into English.

(c) One of the Tocharian A forms, marked with "!" in the table, is written with an error. Find and correct it.

(d) Translate from English: *to know* (A), *he must feel* (A), *they drop* (A), *he felt disgust for the world* (A), *they feel* (A, B), *they collected* (A, B), *he split* (B).

⚠ Tocharian A and B are extinct Indo-European languages from the Tocharian branch, formerly spoken in the Tarim Basin.

Problem 12. Barbareño (2021, MSK)

The words of the Barbareño language are given in singular and plural forms, along with their English translations:

sg.	**pl.**	**trans.**
ʔonyoko	ʔonʔonyokoʔ	shark
ʔixpaniš	ʔixʔixpaʔniš	acorn
ʔawuxa	ʔawʔawuxaʔ	needle
yuxnuc	yuxyuxnucʔ	hummingbird
tomol	tomtoʔmol	boat
takak	taktakakʔ	quail
taniw	tantaʔniw	child
ʔelyeʔwun	ʔelʔelyeʔwuʔn	sword-fish
ʔamut'ey	ʔamʔamut'eʔy	sister
taxama	taxtaxaʔma	skunk
ʔišow	ʔišʔišoʔw	coal
c'inowon	c'inc'inoʔwon	hill
xelex	xelxeʔlex	hawk
puxu	?	dish
ʔališaw	?	day
ʔuwuʔmu	?	food
paxat	?	whale
ʔakayiš	?	bed
qowoč	?	salmon
kawayu	?	horse
q'emen	?	mouse
wuluʔwul	?	lobster

(a) Fill in the plural forms.

⚠ Barbareño is an extinct Chumashan language formerly spoken on the central California coast.

Problem 13. Chakha (2024, MSK)

Some verbs of the Chakha language are given in the 3rd person singular masculine and in a special impersonal form, along with their English translations:

3 sg. m.	**Impersonal**	**Translation**
agbakäbäm	agwakäwim	to stuff
absabäsäm	ab^{w}šab^{w}äšim	to go groping
baräm	b^{w}arim	to speak
g'äbäräm	g'äb^{w}ärim	to pay taxes
kəbäsäsäm	kəwäsäšim	to confuse
näkäbäm	näkawim	to find
täbäsäm	täb^{w}äšim	to fry
ṭəbanäräm	ṭəwanärim	to roll up
abäm	a_im	to give
abetätäm	a_e_ačim	to expand
akäbäräm	akä_ärim	to celebrate
bəṭäbätäm	_ə_ä_ä_im	to stir
gəräbätäm	gərä_ä_im	to turn over
sänäbäm	sänä_im	to spin
täzrabätäm	_äzrab^{w}ačim	to hope